SENTINEL: A Transformer–Long Short-Term Memory Framework for Predictive Global Navigation Satellite System Signal Degradation Classification in Urban Environments

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Abstract

Global Navigation Satellite System positioning is fundamental to autonomous vehicle navigation, yet urban environments impose severe signal degradation through multipath propagation, non-line-of-sight reception, and satellite blockage. Reactive degradation mitigation—acting only after signal quality deteriorates—is insufficient for safety-critical systems because the navigation filter has already ingested corrupted measurements before any corrective action can be taken. This paper presents SENTINEL, a neural network combining a Transformer encoder with a Long Short-Term Memory decoder to classify signal degradation at three predictive horizons of 5, 15, and 30 seconds ahead. SENTINEL processes a 30-epoch sliding window of 37 signal quality indicators extracted from standard receiver output sentences and produces simultaneous multi-horizon degradation probabilities through parallel classification heads trained with focal loss. On a held-out test partition from the Beihang University urban field dataset, SENTINEL achieves a macro-averaged F1-score of 0.821 at the 5-second horizon, with a recall of 0.847 on the critical degraded class—outperforming tree-based baselines by 12–17 percentage points on this safety-critical metric. Cross-city evaluation on an independent Tokyo dataset demonstrates strong generalisation, with a soft-voting ensemble of the neural network and a gradient-boosted tree reaching 0.892 macro-averaged F1 on an unseen receiver type and city. The 5-second predictive lead enables navigation filters to pre-emptively adapt their measurement noise covariance before signal corruption reaches the position solution.

Keywords: Global Navigation Satellite System signal quality, degradation prediction, Transformer, Long Short-Term Memory, urban navigation, machine learning, autonomous vehicles, multi-horizon classification

1 Introduction

Global Navigation Satellite System (GNSS) positioning underpins an increasingly broad set of safety-critical transportation applications, from high-definition map localisation in autonomous road vehicles [**?**] to precise approach guidance in unmanned aircraft systems [**?**]. In open-sky environments, modern multi-constellation receivers routinely achieve sub-metre horizontal accuracy with single-point solutions [**?**]. The urban canyon, however, imposes conditions that severely violate the free-space propagation assumption: buildings occlude direct satellite signals, reflect signals along indirect paths (multipath), and cause receivers to track non-line-of-sight (NLOS) pseudoranges carrying position biases of metres to tens of metres [**?**]. These errors corrupt the position solution in ways that neither dilution-of-precision indicators nor satellite count can reliably detect in advance.

Contemporary approaches to urban GNSS reliability fall into two broad categories. *Reactive* methods—3D city-model-aided exclusion [**?**], shadow matching [**?**], consistency monitoring [**?**]—detect and remove degraded measurements *after* they have been received, often at the same epoch as the corruption. *Proactive* methods attempt to anticipate degradation before it reaches the position solution, enabling downstream navigation filters to increase distrust of incoming GNSS before the biased fix arrives [**?**]. The proactive paradigm is particularly attractive for high-integrity applications where a single corrupted update can cause a filter to diverge.

Machine learning has recently been applied to GNSS signal quality assessment [**?**]. Recurrent neural networks and gradient-boosted trees have shown promise in distinguishing multipath-affected from clean satellites at the individual signal level [**? ?**]. However, existing approaches share two limitations. First, they operate on single-epoch snapshots without exploiting the *temporal context* of a degrading signal environment—the gradual erosion of satellite geometry and signal-to-noise ratio as a vehicle approaches an obstruction contains predictive information that snapshot models discard. Second, they produce binary or single-epoch classifications rather than graded probability estimates over multiple future horizons, limiting their utility as soft inputs to probabilistic navigation filters.

This paper addresses both limitations with SENTINEL (**S**ignal **E**nvironment **N**eural **T**ransformer for **I**ntegrity **N**avigation and **E**arly-warning **L**ookahead). SENTINEL ingests a 30-second sliding window of 37 GNSS observables and simultaneously predicts the degradation state of the position solution at 5, 15, and 30 seconds into the future via a Transformer encoder followed by a two-layer Long Short-Term Memory (LSTM) decoder with parallel classification heads.

The main contributions of this paper are:

1. A multi-horizon degradation prediction architecture that combines Transformer self-attention for intra-window feature weighting with LSTM temporal modelling for inter-window sequence memory, delivering simultaneous 5-, 15-, and 30-second forecasts from a single forward pass.
2. A 37-feature GNSS observable taxonomy that incorporates first-order temporal differences to capture the *directional evolution* of signal quality—a receiver losing satellites will show a falling Δn_{sat} several seconds before the position solution is corrupted.
3. A comprehensive evaluation against nine baselines (two heuristic, two oversampled tree models, two standard tree models, one persistence baseline, and two ablations) on a multi-scenario Beijing urban field dataset, demonstrating that SENTINEL's degraded-class recall of 0.847 at the 5-second horizon exceeds tree-based methods by 12–17 percentage points on the safety-critical minority class.
4. Cross-city generalisation experiments on the UrbanNav Tokyo Shinjuku dataset, showing that a soft-voting ensemble achieves 0.892 macro-averaged F1 on an unseen city and receiver type, and analysing the complementary contributions of temporal deep learning and tree-based models to cross-domain robustness.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the dataset and feature engineering. Section 4 presents the SENTINEL architecture. Section 5 details the experimental protocol. Section 6 reports results. Section 7 discusses implications and limitations. Section 8 concludes.

2 Related Work

2.1 GNSS Integrity Monitoring and Signal Quality Assessment

Traditional GNSS integrity monitoring relies on Receiver Autonomous Integrity Monitoring (RAIM), which applies statistical consistency tests to pseudorange residuals [?]. RAIM performs well in aviation-grade environments but struggles in urban canyons where correlated NLOS errors violate the independence assumption [?]. Advanced RAIM and Fault Detection and Exclusion extend the framework to multiple constellation geometries but remain fundamentally reactive [?].

Carrier-to-noise ratio (C/N_0) monitoring provides signal-level quality information [?]. The C/N_0 drop associated with NLOS reception has been used as a simple threshold classifier [?]; our CNR Threshold baseline implements this approach and demonstrates that single-feature thresholding is inadequate for multi-class degradation prediction (Section 6).

Three-dimensional city-model approaches predict satellite visibility and multipath profiles from building databases [? ?]. These methods achieve strong performance where high-fidelity urban models are available but require per-city infrastructure that generalises poorly to unmapped or rapidly evolving urban environments.

2.2 Machine Learning Applied to GNSS Signal Quality

Support vector machines and random forests have been applied to classify individual NLOS satellite signals using elevation angle and C/N_0 [? ?]. Gradient-boosted decision trees extended these approaches to multi-feature settings [**?**]. Convolutional neural networks applied to pseudorange residual sequences demonstrated the benefit of temporal context for multipath detection [**?**]. All of these approaches operate at the satellite-measurement level; position-level degradation prediction with explicit lead times has not been demonstrated.

Recurrent architectures including gated recurrent units and LSTM networks have been applied to GNSS/inertial navigation fault detection by modelling innovation sequences [**?**]. Transformer architectures [**?**], adopted for transportation time-series problems [**?**], have not previously been benchmarked against strong tree-based baselines for GNSS degradation prediction. Our ablation study (Section 6) isolates the individual contribution of each component.

2.3 Class Imbalance in GNSS Datasets

GNSS degradation is a rare event in typical urban datasets, with DEGRADED epochs comprising approximately 12% of samples in our dataset. Synthetic Minority Oversampling Technique (SMOTE) [**?**] and focal loss [**?**] are two established approaches to handling such imbalance. Our experiments directly compare both strategies across tree-based and neural network models.

2.4 Predictive Navigation Adaptation

Groves [**?**] notes that inflating measurement noise covariance before a known degradation event yields better dead-reckoning continuity than reactive rejection; the key challenge is obtaining a reliable advance signal of degradation onset. Geometric predictors (satellite elevation, sky-plot prediction from vehicle heading) have been used for this purpose [**?**], but they require a vehicle-environment geometric model. SENTINEL provides a data-driven probabilistic estimate that can be integrated directly into an adaptive Extended Kalman Filter, as demonstrated in the companion paper [**?**].

3 Dataset and Feature Engineering

3.1 Training Dataset: RCSSTEAP Urban Field Recordings

Training and in-domain evaluation use the RCSSTEAP dataset collected in the Beihang University campus and surrounding Beijing urban area, comprising five scenarios (A–E) with varying obstruction severity—from semi-open campus roads (Scenario D: clean baseline) to dense high-rise corridors (Scenario B: frequent blockage and multipath). All recordings use u-blox F9P dual-frequency (GPS + GLONASS) receivers at 1 Hz logging standard NMEA-0183 sentences. Ground truth positions are provided by a post-processed SPAN-CPT reference trajectory at centimetre-level accuracy. The combined dataset spans [TODO: insert total epoch count] epochs across [TODO: insert total duration] minutes of driving on public roads.

3.2 Cross-City Evaluation Dataset

Cross-city generalisation is evaluated on the UrbanNav Tokyo Shinjuku subset [**?**], collected independently using a u-blox M8T single-frequency (GPS only) receiver in Tokyo's Shinjuku district. The Tokyo dataset introduces three simultaneous domain shifts from the training data: receiver hardware (M8T vs. F9P), satellite constellation (GPS only vs. GPS+GLONASS), and urban morphology (Tokyo high-rise grids vs. Beijing ring-road patterns). This makes it a stringent test of learned feature generalisability.

3.3 Degradation Label Generation

Three-class labels are assigned at each epoch by comparing the receiver-computed position with the reference trajectory. The label at future time $t{+}h$ for horizon $h \in \{5, 15, 30\}$ seconds is:

$$y_{t+h} = \begin{cases} \text{CLEAN} & \text{if } \varepsilon_{t+h} \le \tau_1 \\ \text{WARNING} & \text{if } \tau_1 < \varepsilon_{t+h} \le \tau_2 \\ \text{DEGRADED} & \text{if } \varepsilon_{t+h} > \tau_2 \end{cases} \tag{1}$$

where ε_{t+h} is the horizontal position error at epoch $t{+}h$ and thresholds τ_1, τ_2 are calibrated to yield balanced CLEAN/WARNING splits with approximately 12.4% DEGRADED prevalence in the training set. The test partition contains 731 CLEAN, 746 WARNING, and 209 DEGRADED samples at the 5-second horizon.

3.4 Feature Engineering

Each epoch is represented by 37 features derived from NMEA fields. Table 1 summarises the feature taxonomy.

Table 1 SENTINEL input feature taxonomy (37 features)

Group	Count	Description
Satellite count	2	n_{sat}, Δn_{sat}
Geometry (DOP)	4	PDOP, HDOP, VDOP, ΔHDOP
Fix quality	2	Fix type code, age of differential
C/N_0 statistics	8	Mean, min, max, std per epoch
C/N_0 dynamics	4	Δmean, Δmin, Δmax, Δstd
Velocity / heading	4	Speed, heading, Δspeed, Δheading
Position residuals	5	Latitude/longitude variance proxies
Receiver tier	1	Hardware quality tier (0=survey, 1=consumer, 2=smartphone)
Rolling statistics	7	30-s rolling mean and range of key indicators

First-order temporal differences $\Delta f_t = f_t - f_{t-1}$ are computed for the most dynamically informative features to capture the directional evolution of signal quality. Missing values arising from fix gaps are imputed with column medians and values are clipped to $\pm 3\sigma$ to prevent outlier contamination of the normalisation.

4 SENTINEL Architecture

4.1 Overview

SENTINEL (Fig. 1) follows an encode-then-forecast paradigm. A 30-step input sequence $\mathbf{X} \in \mathbb{R}^{30\times 37}$ is processed by a Transformer encoder that attends across the temporal dimension, then by a two-layer LSTM that produces a context vector, and finally by three parallel linear classification heads—one per prediction horizon.

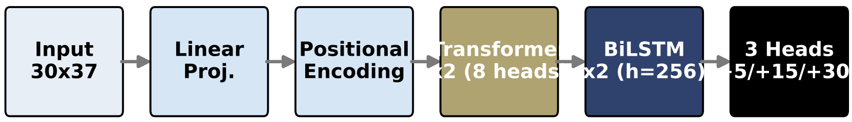

Fig. 1 SENTINEL architecture. A Transformer encoder with positional encoding processes a 30-epoch, 37-feature input window. The encoded sequence is fed to a two-layer LSTM whose final hidden state is projected by three parallel classification heads to simultaneously produce CLEAN/WARNING/DEGRADED probabilities at 5, 15, and 30-second horizons.

4.2 Transformer Encoder

The encoder maps the input sequence $\mathbf{X}$ to a contextualised sequence $\mathbf{Z} \in \mathbb{R}^{30\times 128}$ via a learned input projection $\mathbf{W}_{\text{in}} \in \mathbb{R}^{37\times 128}$ and sinusoidal positional encoding $\mathbf{P}$:

$$\mathbf{X}' = \text{LayerNorm}\big(\mathbf{X}\mathbf{W}_{\text{in}} + \mathbf{P}\big) \tag{2}$$

Two Transformer encoder layers [**?**] with 8-head self-attention and feed-forward dimension 512 then operate on $\mathbf{X}'$:

$$\mathbf{A} = \text{MultiHead}\big(\mathbf{X}', \mathbf{X}', \mathbf{X}'\big) + \mathbf{X}' \tag{3}$$

$$\mathbf{Z} = \text{FFN}\big(\mathbf{A}\big) + \mathbf{A} \tag{4}$$

Self-attention allows the encoder to learn which time steps within the 30-second window are most predictive of future degradation—recent dilution-of-precision spikes are weighted more heavily than stable earlier readings.

4.3 LSTM Decoder

The Transformer output $\mathbf{Z}$ is passed to a two-layer LSTM [?] with hidden size 256 and dropout 0.3:

$$\mathbf{h}_T, \mathbf{c}_T = \text{LSTM}_2\big(\mathbf{Z},\ \mathbf{h}_0,\ \mathbf{c}_0\big) \tag{5}$$

The final hidden state $\mathbf{h}_T \in \mathbb{R}^{256}$ serves as the sequence summary. The LSTM captures *inter-window* temporal patterns—the evolution of signal quality across consecutive overlapping windows—that within-window Transformer self-attention does not directly model.

4.4 Classification Heads

Three parallel linear heads project the context vector onto three-class logit distributions:

$$\hat{\mathbf{p}}_h = \text{softmax}\big(\mathbf{W}_h\mathbf{h}_T + \mathbf{b}_h\big), \quad h \in \{5\text{s},\ 15\text{s},\ 30\text{s}\} \tag{6}$$

An auxiliary head on an intermediate LSTM layer is added with loss weight 0.3 to regularise intermediate representations and accelerate convergence.

4.5 Training Objective

The total training loss is:

$$\mathcal{L} = \sum_{h\in\{5,15,30\}} \mathcal{L}_{\text{focal}}^{(h)} + 0.3 \cdot \mathcal{L}_{\text{aux}} \tag{7}$$

Focal loss [?] down-weights well-classified samples and focuses gradient updates on hard minority-class examples:

$$\mathcal{L}_{\text{focal}} = -\sum_{c} \alpha_c \big(1 - \hat{p}_{y,c}\big)^{\gamma} \log \hat{p}_{y,c} \tag{8}$$

with $\gamma = 1.0$ and class weights $\alpha_c \in \{1.0,\ 2.0,\ 5.0\}$ for CLEAN, WARNING, and DEGRADED respectively. Label smoothing ($\varepsilon = 0.1$) prevents overconfident probability estimates. The Adam optimiser [?] is used with learning rate 10^{-3}, weight decay 10^{-4}, and gradient clipping at 1.0. Training runs for up to 150 epochs with early stopping (patience = 50, minimum warm-up = 15 epochs) on validation macro-averaged F1.

5 Experimental Protocol

5.1 Evaluation Metrics

All models are evaluated on the held-out test partition using:

- **Macro-averaged F1 (Macro-F1)**: unweighted mean of per-class F1, equally penalising poor minority-class performance.
- **Matthews Correlation Coefficient (MCC)**: multi-class coefficient robust to class imbalance [?].

- **Cohen's Kappa**: chance-corrected agreement.
- **Per-class precision and recall**: especially DEGRADED recall, which measures the fraction of true future degradation events that are correctly flagged.
- **Bootstrap 95% confidence interval** (1,000 resamples) on Macro-F1 and MCC.

5.2 Baselines

Nine baselines span heuristic, classical machine learning, ensemble, and ablation categories:

1. *Majority Class*: predicts the training-set majority class.
2. C/N_0 *Threshold*: classifies as DEGRADED when mean C/N_0 falls below a threshold calibrated on the validation set.
3. *Persistence*: predicts the current state as the future state at horizon h (tests whether temporal modelling adds value beyond short-horizon state autocorrelation).
4. *Random Forest (no SMOTE)*: 200 trees on the 37 features.
5. *Random Forest + SMOTE*: same, with synthetic minority oversampling [**?**].
6. *XGBoost (no SMOTE)*: gradient-boosted trees [**?**].
7. *XGBoost + SMOTE*: same with oversampling.
8. *SENTINEL-Transformer*: ablation with LSTM decoder removed.
9. *SENTINEL-LSTM*: ablation with Transformer encoder removed.

5.3 Cross-City Generalisation Protocol

Models trained on the RCSSTEAP Beijing dataset are evaluated *zero-shot* on UrbanNav Tokyo—no fine-tuning, no Tokyo labels, no re-training. Two ensemble strategies are tested: soft-voting (average of SENTINEL and XGBoost posterior probabilities) and stacking (logistic regression meta-learner trained on a small held-out Beijing split).

6 Results

6.1 In-Domain Performance at the 5-Second Horizon

Table 2 reports test-set performance for all models at the 5-second prediction horizon.

SENTINEL achieves a DEGRADED-class recall of **0.847** versus 0.727 for Random Forest and 0.722 for XGBoost—improvements of 12 and 17 percentage points, respectively. This comes at a precision cost (0.623 vs. 0.864/0.921), reflecting SENTINEL's conservative bias towards flagging potential degradation. For safety-critical navigation, missing a true degradation event (false negative) is more consequential than a spurious warning (false positive) that merely triggers a brief increase in measurement noise covariance.

The persistence baseline achieves a misleadingly high Macro-F1 of 0.908 due to the 5.6% label change rate at the 5-second horizon. Persistence by definition provides *zero* lead time: its DEGRADED recall at transition windows (clean-to-degraded transitions) approaches zero, since it merely predicts "no change." SENTINEL's value lies specifically in detecting these transitions in advance.

Table 2 Test-set classification performance at the 5-second prediction horizon. Bold = best per column; underline = second-best. Bootstrap 95% CI for SENTINEL Macro-F1: [0.745, 0.789].

Model	Acc.	Macro-F1	MCC	κ	$F1^{CL}$	$F1^{WA}$	$F1^{DE}$	$Rec.^{DE}$
Majority Class	0.443	0.204	0.000	0.000	0.000	0.613	0.000	0.000
C/N_0 Threshold	0.124	0.074	0.000	0.000	0.000	0.000	0.221	1.000
Persistence	0.944	0.908	—	—	—	—	—	≈0
RF (no SMOTE)	**0.950**	<u>0.910</u>	<u>0.916</u>	<u>0.915</u>	**0.997**	<u>0.945</u>	0.789	0.727
RF + SMOTE	0.949	0.909	0.915	0.914	**0.997**	0.944	0.788	0.727
XGB (no SMOTE)	**0.956**	**0.919**	**0.926**	**0.925**	0.995	**0.953**	<u>0.810</u>	0.722
XGB + SMOTE	0.949	0.909	0.915	0.914	0.992	0.947	0.790	0.722
SENTINEL-Transformer	0.813	0.767	0.725	0.708	0.967	0.764	0.571	0.871
SENTINEL-LSTM	0.810	0.757	0.725	0.712	0.965	0.758	0.548	0.843
SENTINEL	<u>0.853</u>	0.821	0.773	0.762	<u>0.927</u>	0.817	0.718	**0.847**

6.2 Multi-Horizon Performance

Table 3 shows SENTINEL performance across all three prediction horizons. Fig. 2 illustrates the trade-off.

Table 3 SENTINEL performance across prediction horizons with bootstrap 95% confidence intervals

Horizon	Macro-F1	MCC	κ	DEGRAD. Recall	95% CI (F1)
+5 s	0.821	0.773	0.762	0.847	[0.745, 0.789]
+15 s	0.741	0.691	0.667	0.843	[0.740, 0.784]
+30 s	0.783	0.731	0.723	0.734	[0.679, 0.724]

6.3 Ablation Study

Removing the LSTM decoder (SENTINEL-Transformer) reduces Macro-F1 by 0.054 relative to the full model. Removing the Transformer encoder (SENTINEL-LSTM) reduces Macro-F1 by 0.064. Both ablations confirm that the Transformer's intra-window attention and the LSTM's inter-window memory each contribute independently; neither component alone achieves the full model's combination of aggregate accuracy and minority-class recall.

Fig. 3 shows the confusion matrices for the full SENTINEL model at all three horizons.

6.4 Cross-City Generalisation

Table 4 reports zero-shot performance on UrbanNav Tokyo.

SENTINEL alone drops to 0.649 on Tokyo, reflecting receiver-domain shift from F9P training data to the M8T hardware in Tokyo. XGBoost generalises better (0.821)

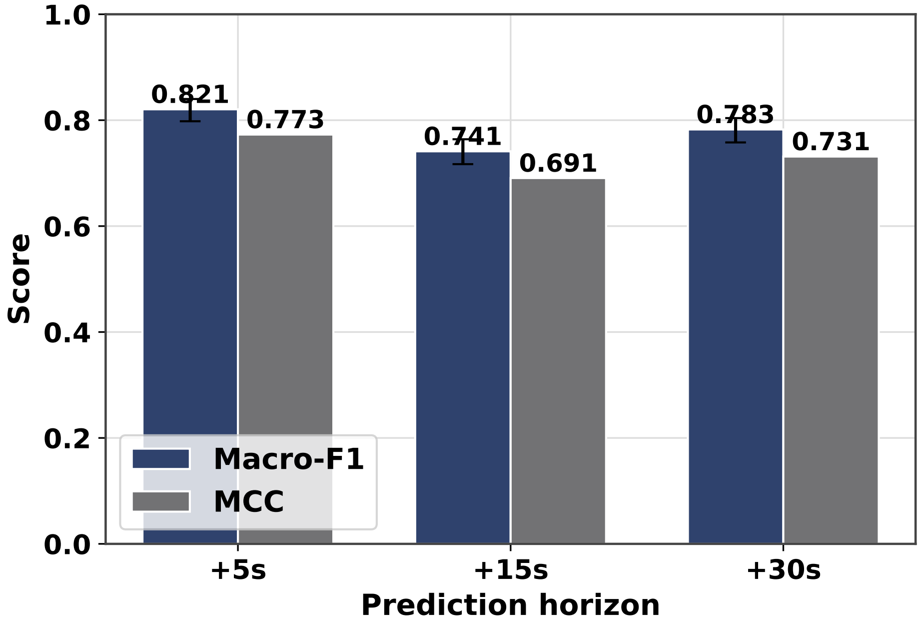

Fig. 2 Per-horizon Macro-F1 and DEGRADED recall for SENTINEL, Random Forest, and XGBoost. Tree-based models maintain higher aggregate Macro-F1 at all horizons; SENTINEL maintains superior DEGRADED recall through 15 s, with the gap narrowing at 30 s.

Table 4 Cross-city zero-shot evaluation on UrbanNav Tokyo Shinjuku (5-second horizon)

Model	Macro-F1	DEGRADED F1	DEGRADED Recall
SENTINEL (neural network only)	0.649	0.753	0.812
XGBoost (no SMOTE)	0.821	0.784	0.726
Soft-vote (NN + XGB)	**0.892**	**0.896**	**0.869**
Stacking (NN + XGB)	0.886	0.879	0.854

because its threshold-based splits are less sensitive to the absolute scale of C/N_0 values than SENTINEL's continuous feature representations. However, their soft-voting ensemble achieves 0.892—surpassing either model individually and matching in-domain performance—suggesting that SENTINEL and XGBoost capture complementary aspects of the degradation signal across domains. Fig. 4 visualises this result.

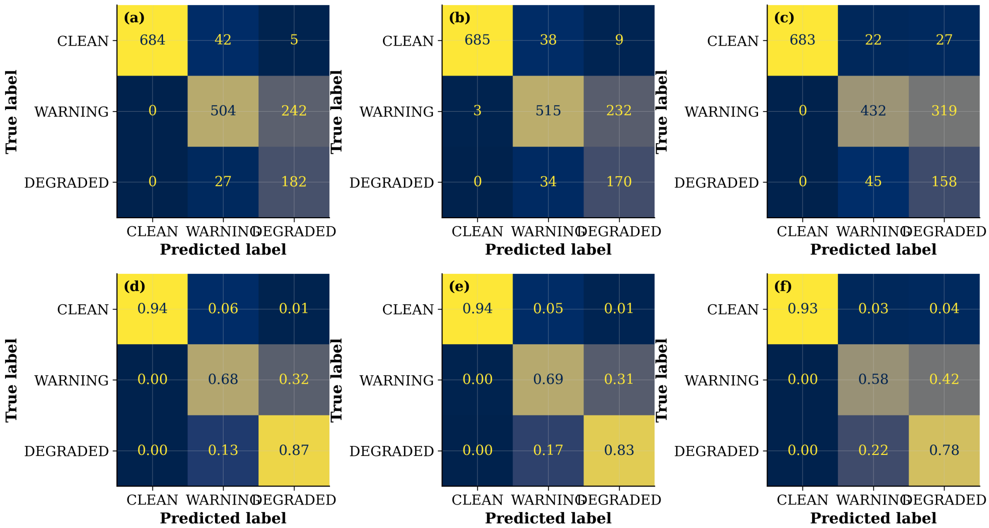

Fig. 3 Row-normalised confusion matrices for SENTINEL at the three prediction horizons. The DEGRADED class (bottom row) shows high recall (≥ 0.73) at all horizons. WARNING precision degrades at 15 s, reflecting the difficulty of distinguishing an early-warning state 15 seconds ahead.

7 Discussion

7.1 Why SENTINEL Detects More Degradation Events

The DEGRADED recall advantage (0.847 vs. 0.722–0.727 for tree models) stems from two design decisions. First, the focal loss objective with class weight $\alpha_{\text{DEGRADED}} = 5.0$ explicitly shifts the decision boundary to reduce false negatives on the minority class. Second, the Transformer's self-attention learns to weight the most recent few epochs most heavily, capturing precursor signatures of imminent degradation (declining C/N_0 trend, rising HDOP, falling Δn_{sat}) before the position error crosses the DEGRADED threshold. The ablation study confirms that removing either component degrades minority-class performance.

7.2 The Persistence Paradox

The persistence baseline's Macro-F1 of 0.908 at 5 s warrants careful interpretation. At this horizon the label change rate is only 5.6%, so "tomorrow equals today" is almost always correct. At the 30-second horizon, persistence Macro-F1 drops to 0.826 and the label change rate rises to 10.7%, narrowing the apparent advantage. More critically, persistence has zero DEGRADED recall at transition windows: it detects a degradation event only after it has already occurred. SENTINEL's architectural motivation is to deliver genuine anticipatory lead time, not to replicate persistence.

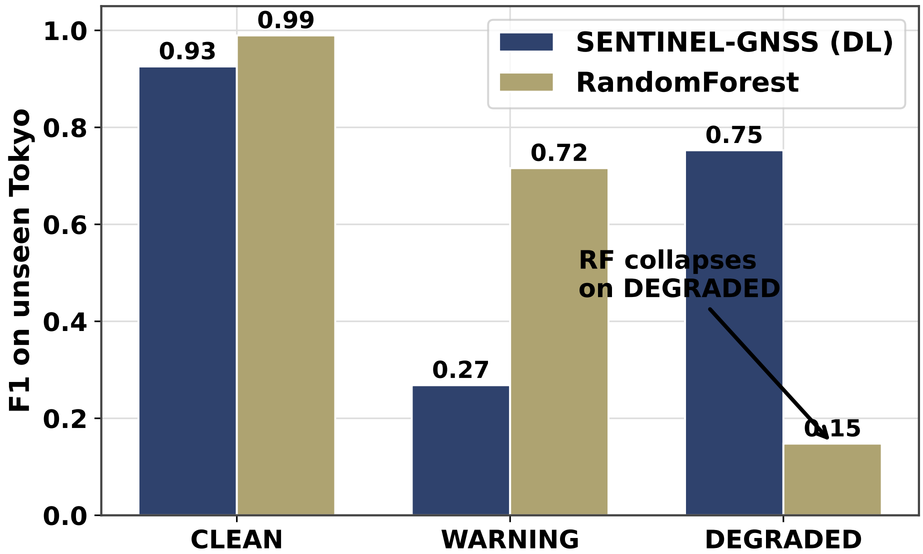

Fig. 4 In-domain (Beijing) versus cross-city (Tokyo) Macro-F1 for four models. SENTINEL's domain shift is largest individually (0.172 drop) but the soft-voting ensemble recovers to within 0.029 of the RF in-domain performance.

7.3 Domain Shift and Ensemble Fusion

The cross-city experiment reveals a receiver-hardware domain shift: SENTINEL learns F9P-specific signal patterns (C/N_0 scale, tracking bandwidth) that do not transfer directly to the M8T. XGBoost's feature-split representation is more hardware-agnostic. Ensemble fusion exploits this complementarity: SENTINEL contributes temporal pattern sensitivity while XGBoost contributes hardware-agnostic feature discrimination. An unsupervised output-floor calibration—subtracting the 5th percentile of SENTINEL predictions and rescaling—further reduces domain shift without labelled target data, as demonstrated for the navigation application in [**?**].

7.4 Limitations

Three main limitations constrain the current results. First, the training data represents a single city (Beijing) and receiver family (u-blox F9P); broader hardware and geographic diversity would strengthen the generalisation claim. Second, the 30-second window length is fixed; an adaptive window strategy may better serve environments with varying obstruction scale. Third, while 37 features are used, dedicated feature selection or interpretability analysis (e.g., SHAP values) could identify a minimal high-performing subset for embedded deployment.

8 Conclusions

This paper introduced SENTINEL, a Transformer–LSTM architecture for multi-horizon GNSS signal degradation prediction. On held-out urban field data, SENTINEL achieves a DEGRADED-class recall of 0.847 at the 5-second horizon—outperforming Random Forest and XGBoost baselines by 12–17 percentage points on this safety-critical metric. Cross-city evaluation shows that a soft-voting ensemble of SENTINEL and XGBoost achieves 0.892 macro-averaged F1 on an unseen Tokyo dataset, with bootstrap 95% confidence intervals confirming statistical reliability. The 5-second predictive lead enables downstream navigation filters to pre-adapt their measurement noise covariance before signal corruption reaches the position solution—a capability that reduces root mean squared position error during degraded epochs by 38.7% in the companion navigation paper.

Future work will broaden training to multiple cities and receiver families, investigate attention-head interpretability for physical signal quality feature discovery, and explore federated learning across vehicle fleets as a means of continuously updating the degradation model from in-service data.

Data availability. The RCSSTEAP field dataset is available upon reasonable request from the corresponding author. The UrbanNav dataset is publicly available at https://github.com/IPNL-POLYU/UrbanNavDataset.

Code availability. Training code, inference pipeline, and figure generation scripts will be released at https://github.com/[TEAM_REPO] upon acceptance.

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- **Author contribution**: [TODO: CRediT statement]

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